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THEORY OF INCOME AND EMPLOYMENT

QUICK REVISION

1. Aggregate Demand:
(i) Aggregate demand refers to the total expenditure that the residents of a country are ready to incur on the purchase of goods and services, given their level of income.
(ii) Components:
(a) Consumption demand (C),
(b) Investment demand (I),
(c) Government's demand for goods and services (G) and
(d) Net demand by foreigners Or Net Exports (X-M)
There are only two components of AD in a two sector economy that is consumption demand (C) and investment demand (1). Thus, AD = C + I.
(ii) Nature of AD: AD depends upon the level of income. Higher the level of income, higher will be the level of AD. AD curve is upward sloping.

2. Aggregate Supply:
(i) Aggregate supply refers to the total output of goods and services available for purchase by the economy during a period of one year.
(ii) Components: Aggregate supply is the sum of consumption expenditure (C) and savings (S).Thus, AS = C+S.
(ii) Nature of AS: Aggregate Supply (or flow of goods and services in the economy) increases or decreases in proportion to increase in employment. As line is 45° line which implies identify between Y on the X-axis and C+S on the Y-axis.

3. Consumption Function:
(i) The functional relationship between consumption and income is called consumption function. C = f(Y)
(ii) Keynes's Psychological Law of Consumption: This law states that as income of the people rises, their consumption also rises but increase in consumption is less than increase in income.
(iii) Measures of Propensity of Consume: (a) Average Propensity to Consume (APC) and (b) Marginal Propensity to Consume (MPC)
APC = C/Y; MPC = AC/AY
APC is the value of consumption function at a particular level of national income. Marginal propensity to consume measures change in consumption per unit change in income.

4. Saving Function:
(i) It expresses the functional relationship between savings and income and is given by:
S = f(Y)
(ii) Two Measures of Propensity to Save: (a) Average propensity to save (APS) and (b) Marginal propensity to save (MPS). Average propensity to save is the ratio of total saving to total income. Marginal propensity to save in the ratio between changes in saving to change in income.
(iii) Negative Savings or Dissavings: It refers to a phenomenon when consumption exceeds income.
(iv) Slope: The slope of saving function is given by 1-b where b denotes marginal propensity to consume.

5. Determination of Equilibrium Level of Income and Employment: Equilibrium level is that level of income where aggregate demand equals to aggregate supply and the level of planned savings equals to planned investment. Thus, there are two approaches to determine the level of equilibrium:
(i) AD-AS Approach: Equilibrium level of output (income) is determined when quantity of output produced (AS) is equal to quantity demanded (AD). That is in equilibrium AD = AS.
(ii) S-1 Approach: The equilibrium level of income is determined at a level when planned savings (S) equal to planned investment (1), i.e., S = 1.
When aggregate demand is equal to the aggregate supply at full employment level, it is called full employment equilibrium.
When aggregate demand is equal to the aggregate supply at a level where the resources are not fully employed, it is called under employment equilibrium.
When AD is equal to AS beyond the full employment level it is called over full employment equilibrium.
Multiplier:
(i) Multiplier (K) refers to the ratio of change in income (AY), to a change in investment (AI). $K = \Delta Y / \Delta I$. The minimum value of multiplier can be one and the maximum value can be infinity.
(ii) Multiplier is directly related with the MPC, i.e., K = 1/1 - MPC.
(iii) Multiplier is inversely related with the MPS, K = 1/MPS.

6. The Concepts of Full Employment and Involuntary Unemployment: Full employment refers to a situation in which all able and willing persons are able to find jobs at given wage rate.
In a situation of full employment, frictional unemployment and structural unemployment can exist. If some able persons are unemployed against their wishes they are called involuntarily unemployed.

EXCESS AND DEFICIENT DEMAND

1. Situation of Demand: There are three situations of demand. These are: (i) Deficient demand, (ii) Excess demand and (ii) Sufficient demand.
2. Deficient Demand:
(i) Meaning: Deficient demand is a situation which arises when AD is less than the AS corresponding to the full employment in the economy. This gives rise to a deflationary gap.
(ii) Causes: Deficient demand may be caused by the following factors:
(a) Credit contraction (b) Cut in Government expenditure (c) Rise in taxes (d) Rise in propensity to save.
(iii) Consequences:
(a) Inventory stock of unsold goods will pile up with the producers beyond the desired level.

(b) Producers would plan to cut in production in next round.
(c) Reduction in planned output would cause reduction in the level of income and employment.

(iv) Remedial Measures:

Fiscal Measures: (a) Taxes should be reduced. (b) Government should send more on public works like construction of roads, bridges, buildings canals, education, health, etc.

Monetary Measures: (a) The central bank should reduce the bank rate and thus making the credit cheaper. (b) The central bank should buy Government securities from the open market and thus enhance the money supply in the economy. (c) Reduction in margin requirement of loan to encourage borrowings.

3. Excess Demand:

(i) Excess demand refers to a situation when AD in the economy is more than AS at full employment level of Output.

(ii) Causes: Excess demand in an economy may be caused by any one of the following factor: (a) Increase in money supply (b) Rise in propensity to consume (c) Increase in Government expenditure (d) Cut in taxes (e) Increase in exports.

(iii) Consequences: When aggregate demand increases beyond full employment level, price level rises which implies a situation of inflation.

(iv) Remedial Measures:

Fiscal Measures: (a) Taxes should be increased. (b) Government expenditure should be reduced.

Monetary Measures: (a) Reduction in availability of credit through: (1) increase in bank rate. (II) increase in legal reserve requirements.

1 MARK QUESTIONS

A. FILL IN THE BLANKS

1. The slope of the consumption function is called _____ .
2. The slope of saving curve is called _____ .
3. The sum of APC and APS is equal to _____ .
4. At break-even point savings are _____ .
5. The investment function gives the relationship between investment and the _____ .
6. Net export is the difference between _____ and _____ .
7. If MPC is 0.8 and Y = 100, C would be, _____ .
8. If MPS is zero, MPC would be _____ .
9. If consumption is 100 and income is 200, APS = _____ .
10. If savings are 250 and income is 1000, APC = _____ .
11. At equilibrium level of income _____ is equal to planned investment.
12. The functional relationship between _____ and _____ is known as consumption function.
13. AS is a _____ line.
14. Aggregate supply is _____ + _____ .
15. AD curve starts from _____ .
16. If MPS is 0.2, the value of multiplier is _____ .
17. Raising taxes would help in correcting _____ demand.
18. Consumption and _____ are directly related to each other.
19. Multiplier is the ratio of change in income to the change in _____ .

20. If MPS is 0.1, the multiplier = _____ .
21. Consumption done at zero level of income is known as _____ .
22. Investment done with the profit motive is known as _____ .
23. There is a negative relationship between _____ and multiplier.
24. APS can never be 1 or more than 1 because saving can never be _____ or the national income.
25. Autonomous investment curve is a _____ line.
26. Y = _____ .
27. AD= _____ .
28. AS= _____ .
29. APC = _____ .
30. APS = _____ .
31. MPC = _____ .
32. MPS = _____ .
33. Multiplier K = _____ , _____ ,_____ .
34. EQUILIBRIUM = AD = _____ , S = _____ .

ANSWERS

1. MPC	2. MPS	3. One	4. Zero
5. Rate of interest	6. Exports and imports	7. 80	8. One
9. 0.5	10. 0.75	11. Planned saving	12. Consumption and Income
13. 45 degree	14. Consumption, Saving Income	15. Above the origin	16. 5
17. Excess	18. Income	19. Investment	20. 10
21. Autonomous consumption	22. Induced investment	23. MPS	24. Equal to or more than
25. Horizontal straight	26. C+S	27. C+1	28. C+S
29. C/Y	30. S/Y	31. AC/AY	32. AS/AY
33. $\frac{\Delta Y}{\Delta I}, \frac{1}{1-MPC}, \frac{1}{MP}$	34. AS, I		

B. TRUE OR FALSE

1. The value of APS can never be more than national income.
Ans. [True] Because saving can never be more than national income.

2. Sum of APC and MPC is always equal to one.
Ans. [False] Because APC is the ratio of consumption and income and MPC is the ratio of change in consumption to the change in income.

3. Value of APS can never be less than zero.
Ans. [False] Value of APS can be less than zero when consumption is more than income.

4. Value of investment multiplier varies between zero and infinity.

Ans. [True] If MPC = 0, then K will be 1 and if MPC-1 then value of K will be infinity (**).

5. Saving and investment are always equal.
Ans. [False] Because savers and investors are different people who save and invest with different motives.

6. An economy is only in equilibrium at full employment level.
Ans. [False] As equilibrium can occur with full employment, under employment and over full employment.

7. Inflationary gap is measured by deficient demand.
Ans. [False] Because inflationary gap is measured by excess demand.

8. Consumption is positively related to income but (C) is not zero when (Y) is zero.
Ans. [True] C is positively related to Y but at zero level of income, a minimum level of consumption is necessary for survival.

9. Break-even point is struck when saving is equal to zero.
Ans. [True] Because at break-even point, Consumption = Income so, saving = 0.

10. Unemployment equilibrium indicates excess capacity in the economy.
Ans. [True] Because AD is less than what is necessary for full employment of resources implying existence of excess capacity

11. Deflationary gap can be corrected by increasing the level of AD.
Ans. [True] As it is the deficiency of AD which causes deflationary gap.

12. Aggregate demand depends upon level of income.
Ans. [False] Aggregate demand depends upon level of income.

13. APC is never negative.
Ans. [True] Because there must be some consumption at each level of income.

14. Aggregate demand = C + S.
Ans. [False] Because AD = C + I and AS = C + S.

15. Investment multiplier is directly related to MPC.
Ans. [True] Because with more consumption demand increases which increases the production and investment in the Economy.

C. MULTIPLE CHOICE QUESTIONS

AGGREGATE DEMAND, AGGREGATE SUPPLY AND RELATED CONCEPTS

1. What are main components of aggregate demand in an open economy?
(a) Consumption
(b) Investment

(c) Consumption and investment
(d) Consumption + Investment + Net Export + Government expenditure

2. Which of the following statement is true?
(a) MPC = $\frac{C}{Y}$
(b) MPC = $\frac{\Delta S}{\Delta Y}$
(c) MPC = $\frac{\Delta C}{\Delta Y}$
(d) None of these

3. APC+APS=?
(a) 0
(b) 1
(c) ∞
(d) None of these

4. In a closed economy aggregate demand equals:
(a) C+ I
(b) C+I+G
(c) C+ I + IX +G
(d) None of these

5. MPC+ MPS-?
(a) O
(b) 1
(c)∞
(d) None of these

6. How do we calculate average propensity to consume?
(a) $\frac{C}{Y}$
(b) $\frac{\Delta C}{\Delta Y}$
(c) $\frac{S}{Y}$
(d) $\frac{\Delta C}{\Delta Y}$

7. How can we calculate average propensity to save ?
(a)$\frac{\triangle s}{\triangle y}$
(b)$\frac{s}{y}$
(c)$\frac{\triangle c}{\triangle y}$
(d)$\frac{c}{y}$

8. How do we calculate MPS?

(a)$\frac{\triangle s}{\triangle y}$

(b)$\frac{\triangle c}{\triangle y}$

(c)$\frac{s}{y}$

(d)$\frac{s}{I}$

9. Which of the following equation is true to calculate aggregate supply?

(a) AS = C + I

(b) AS = C + S

(c)AS = S + I

(d)None of these

10. When MPC=0.8, what would be MPS?

(a) -0.2

(b) 0.2

(c) -1.5

(d) None of these

11. When APS = -0.5, what would be APC?

(a) 0.5

(b) 1.5

(c) -1.5

(d) None of these

12. From where aggregate demand curve starts?

(a) From origin

(b) From X-axis

(c) From Y-axis

(d) none of these

13. From where aggregate supply curve starts?

(a) From X-axis

(b) From Y-axis

(c)From origin

(d)None of these

14. When income is 500 crores and consumption is 400 crores, what would be APC?

(a) 1.25

(b) 0.8

(c)0.2

(d) None of these

15. If APC is 60%, what would be APS?
(a) 0.4
(b) 1.6
(c)0.2
(d)None of these

16. Which of the following equation is true for saving function?
(a)S= f(C)
(b) S = f(Q)
(c) S=f(Y)
(d) None of these

17. Which is main determinant of consumption?
(a) Production level
(b) Income level
(c) Employment level
(d) Any of these

18. Which of the following is an equation for consumption?
a) C = f(S)
(b) C = f(Q)
(c) S = f(Y)
(d) None of these

19. When APC = 1, what would be APS?
(a) 0
(b) -1
(c)2
(d) None of these

20. What type of curve autonomous investment has?
(a) Parallel to X-axis
(b) Starts from origin
(c) Parallel to Y-axis
(d) None of these

21. What motive induced investment has?
(a) Public welfare
(b)Profit earning
(c) Development of public sector
(d) None of these

22. What type of curve induced investment has?
(a) Slope upward from left to right
(b) Parallel to X-axis
(c) Slopes downward from left to right
(d) Parallel to Y-axis

23. Aggregate supply is equal to:
(a) level of income
(b) level of employment
(c) aggregate expenditure
(d) all of these

24. Which of the following is true?
(a) MPC + MPS = 0
(b) MPC + MPS <1
(c) MPS + MPC > 1
(d) MPC + MPS = 1

25. If MPS is 30%, what would be MPC?
(a) 70%
(b) 130%
(c) 60%
(d) None of these

ANSWERS

1.d	2.c	3.b	4.b	5.b	6.a	7.b	8.a	9.b	10.b
11.b	12.c	13.c	14.b	15.a	16.c	17.a	18.c	19.a	20.a
21.b	22.a	23.a	24.d	25.a					

DETERMINATION OF EQUILIBRIUM LEVEL OF INCOME AND EMPLOYMENT

1. Which of the following is true in case of classical theory of employment?
(a) Economy remains in full employment level
(b) No possibility of over production or under production
(c) Both (a) and (b)
(d) None of the above

2. There is full employment in an economy, when there exists no:
(a) involuntary unemployment
(b)voluntary employment
(c) disguised unemployment
(d) None of these

3. An economy is in equilibrium when:
(a) AD = AS
(b) saving = investment
(c)both (a) and (d)
(d) None of these

4. In the situation of equilibrium of an economy, there is:
(a) AS = C + I
(b) C+S=C+I
(c) S=I
(d) All of these

5. When AD equals AS at less than full employment level, it is known as:
(a) under employment level
(b) over full employment level
(c) full employment level
(d) any of these

6. An increase in employment level depends upon increase:
(a) consumption level alone
(b) saving level alone
(c) on aggregate demand
(d) none of these

7. When AD is not equal to AS, which force brings them back in equilibrium?
(a) Level of investment
(b) Level of income
(c)Level of employment
(d) Any of these

8. Generally an economy is in equilibrium at which of the following level?
(a) Under employment
(b) Full employment
(c) over full employment
(d) None of these

9. Which of the following is true?
(a) An economy is always in full employment level
(b)APC + MPC = 1
(c) APS + MPS = 1
(d) An economy has generally under employment level

10. Which of the following is not a component of aggregate demand in closed economy?
(a) Aggregate consumption
(b) Aggregate investment
(c) Net exports
(d) public investment

11. Which of the following equation is true?
(a) C+1=C+S
(b) MPC = MPS
(c) AD = C + S
(d) AS = C + M

12. Employment theory is related to which of the following theory?
(a) Static economics
(b) Micro economics
(c) Macro economics
(d) None of these

ANSWERS

1.c	2.a	3.c	4.d	5.a	6.c	7.b	8.a	9.d	10.c
11.a	12.c								

INVESTMENT MULTIPLIER AND ITS WORKING

1. Which of the following formula is correct?
(a) K =$\frac{1}{MPC}$
(b) K =$\frac{1}{MPS}$
(c) K =$\frac{1}{1-APC}$
(d) K =$\frac{1}{MPC-1}$

2. How multiplier can be expressed?
(a) K=$\frac{\triangle Y}{\triangle I}$
(b) K=I-S
(c)K=$\frac{S}{I}$
(d None of these

3. When MPC 1, multiplier would be:
(a) K=0
(b)K=∞
(c)K=1
(d)between 0and 1

4. If MPC = 0, the K would be:
(a) 1
(b)infinity
(c)0
(d)between 0 and 1

5. Multiplier varies between .....
(a) 1 to 0
(b)1 to ∞
(c)∞ to 0
(d) None of these

6. If MPC is 0.8, multiplier would be:
(a) 5
(b) 4
(c) 3
(d) none of these

7. When MPS = 0.2, what would be the size of multiplier?
(a) 5
(b) 2
(c) 20
(d) None of these

8. When MPS is 0.1 and investment is 100 crs, what would be the increase in income?
(a) 10 crs
(b)1000crs
(c)100crs
(d)None of these

9. Investment multiplier depends upon which of the following factor?
(a)MPC
(b)MPS
(c) both (a) and (b)
(d) None of these

10. With an 50 crs investment income rises to 250 crs. What would be size of multiplier?
(a) 5
(b)10
(c)3
(d) None of these

11. If MPC-MPS, what would be size of multiplier?
(a)5
(b)2
(c)1
(d)None of these

12. What relationship exists between size of MPC and level of multiplier?
(a) Direct
(b) Inverse
(c) Any of these
(d) None of these

13. How MPS and multiplier related?
(a) Inversely
(b) Directly
(c)Any of these
(d) None of these

14. When MPS is zero, what is the size of K?
(a) 1
(b) ∞
(c) 2
(d) None of these

15. Multiplier establishes relationship between which of the following?
(a) Investment and income
(b) Income and consumption
(c) Saving and investment
(d) None of these

ANSWERS

1.b	2.a	3.b	4.a	5.b	6.a	7.a	8.b	9.c	10.a
11.b	12.a	13.a	14.b	15.a					

PROBLEM OF EXCESS DEMAND AND DEFICIENT DEMAND

1. Problem of excess demand arises on which of following level:
(a) Full employment
(b) Under employment
(c) Over full employment
(d) None of these

2. The excess demand arises, if:
(a) AD = AS
(b) AD <AS
(c) AD > AS on fully employment level
(d) None of these

3. On which level of employment problem of excess demand arises?
(a) Inflation
(b) Deflation
(c)state of depression
(d)None of these

4. On which level of employment problem of deficient demand arises?
(a) Full employment
(b) under employment
(c) Over full employment level
(d)None of these

5. What problem is created due to deficient demand?
(a) Inflation
(b) Deflation
(c) Boom condition
(d)none of these

6. When the problem of excess demand arises?
(a) AD <AS
(b)AD>AS
(c) AD > AS as full employment demand
(d)None of these

7. What is the main reason for excess demand?
(a) Increase in exports
(b) Deficit financing
(c) Increase in investment
(d) All of these

8. How inflationary tendency can be measured?
(a) With deficient demand
(b) With excess demand
(c) Either of these
(d) None of these

9. Which of the following is main reason of deficient demand?
(a) Less consumption expenditure
(b) Low exports
(c) Surplus budget
(b) Low exports

10. How deflationary gap can be measured?
(a) With deficient demand
(b) With excess demand
(c) Either of these
(d) None of these

11. What are economic indicators of excess demand?
(a) Depression
(b) Slump
(c) Prosperity
(d) None of these

12. What are economic indicators of Deficient demand?
(a) Boom conditions
(b) Inflation
(c) Depression
(d) Prosperity

13. In which situation level of income and employment rises?
(a) Under deflationary situation
(b) Under excess demand
(c) Under deficient demand
(d) None of these

14. What types of problem is created when there is excessive rise in money supply?
(a) Excessive demand
(b) Deficient demand
(c) Under deficient demand
(d) None of these

15. Excessive rise in public expenditure creates which of following problem?
(a) Excess demand
(b) Deficient demand
(c) Deflationary tendency
(d) Any of these

16. Which of following is found, normally in an economy?
(a) Deflationary
(b) Inflationary
(c) Both of these
(d) None of these

17. Which of the following statement is true?
(a) Deficient demand gives birth to inflation
(b) Excess demand gives birth to deflation
(c) Excess demand is a measure of inflation
(d) None of these

18. Which of following creates depression ?
(a) Excess demand
(c) Constantly deficient demand
(b) Deficient demand
(d) Any of these

ANSWERS

1.c	2.c	3.a	4.b	5.b	6.c	7.d	8.b	9.d	10.a
11.c	12.c	13.b	14.a	15.a	16.c	17.c	18.c		

MEASURE TO CORRECT EXCESS AND DEFICIENT DEMAND

1. Which of the following is a fiscal measure?
(a) Public expenditure
(b) Taxation
(c) Public debt
(d) All of these

2. Which of the following problem can be solved by reducing deficit financing?
(a) Excess demand
(b) Deflationary gap
(c) Deficient demand
(d) All of these

3. Which problem can be solved by reducing tax rates?
(a) Excess demand
(b) Inflationary gap
(c) Deficient demand
(d) Any of these

4. By reducing public expenditure, which of following problem can be solved?
(a) Excess demand
(b) Deficient demand
(c) Deflationary gap
(d) Depression

5. Which of the following problem can be solved by raising debts?
(a) Depression
(b)Deficient demond
(c)Excess demond
(d)Any of these

6. An increase in taxation is a best measure for which of following situation?
(a) Excess demand
(b) deficient demand
(c)Depression
(d)None of these

7. Which of the following is a best measure to contain excess demand?
(a) Increase in taxation
(b) Increase in public debts
(c) Both of these
(d)None of these

8. Cheap money policy is the best measure to solve the problem of:
(a) deflation
(b) depression
(c) deficient demand
(d)All of these

9. Decrease in cash reserve ration and repo-rate is a best solution to solve the problem of:
(a) deficient demand
(b) depression
(c) both of these
(d)None of these

10. Which of the following measure is most suitable to solve the problem of deficient demand?
(a) Fall in cash reserve ratio
(b) Increase in public expenditure
(c) Low repo-rate
(d) Any of these

11. Fall in repo-rate and reserve repo-rate solves the problem
(a) deficient demand
(b) depression
(c) deflation
(d) all of these

12. Sale of securities in open market solve the problem of:
(a) deficient demand
(b) depression
(c) deflation
(d) all of these

13. Which of following measure should be adopted to control deficient demand?
(a) Rise to public expenditure
(b) Fall in bank rate
(c) Fall in tax rates
(d) Any of these

14. Which of the following measure should be adopted to remove depression?
(a) Raise bank rate and tax rates
(b) To raise marginal
(c) Fall in tax rates
(d) All of these

15. Purchase of securities from open market is useful to remove the problem of:
(a) deficient demand
(b) excess demand
(c) inflationary tendency
(d) any of these

16. Which of following statement is true?
(a) Fall in tax rates can be useful to control excess demand.
(b) Purchase of securities from open market is useful to control excess demand.
(c) Deficit financing is useful to control deficient demand.
(d) All of these

17. A fall in cash reserve ratio and statutory liquidity ratio is useful when there is:
(a) excess demand
(b) deficient demand
(c) inflationary situation
(d) any of these

ANSWERS

1.d	2.a	3.c	4.a	5.c	6.a	7.c	8.d	9.c	10.d
11.d	12.b	13.d	14.b	15.a	16.c	17.b			

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : APS can never be 1 or more than 1.
Reason (R) : Saving can never be equal to or more than national income.

2. Assertion (A) : APC falls with increase in income.
Reason (R) : As the income increases, the proportion of income spent on consumption falls.

3. Assertion (A) : Autonomous investment curve is a horizontal straight line parallel to X-axis
Reason (R) : Autonomous investment is directly related to the national income.

4. Assertion (A) : There is a direct relation between value of multiplier and MPC
Reason (R) : One man's income is other man's expenditure.

5. Assertion (A) : Aggregate supply is known as the national income of the country.
Reason (R) : Aggregate supply is a flow of goods and services produced in a country during a given period of time.

ANSWERS

1.c	2.c	3.a	4.c	5.c

2 MARK QUESTIONS

1. What is aggregate demand? Define it.
Ans. Aggregate demand refers to the total demand for all goods and services in an economy.

2. What are the components of AD?
Ans. Components of AD: (i) Private consumption demand, (ii) Private investment demand, (iii) Demand for goods and services by the Government and (iv) Net exports.

3. What is aggregate supply? Define it.
Ans. Aggregate supply refers to the total quantity of goods and services produced by all the producers in an economy.

4. What type of relation is found between income and consumption?

Ans. Consumption increases with the increase in income but less than the increase in income. C = f(Y).

5. Define marginal propensity to save.
Ans. MPS is the ratio of the change in savings to the change in income. MPS = $\triangle S/\triangle Y$.

6. Define MPC.
Ans. MPC is the ratio of change in consumption to the change in income. MPC = $\triangle C/\triangle Y$.

7. What determine the level of household consumption demand in an economy?
Ans. The level of disposable income and propensity to consume determine the household consumption demand in an economy. C=f (Y).

8. Define propensity to consume.
Ans. The functional relationship between consumption and income is known as propensity to consume. C = f(Y).

9. What is average propensity to consume?
Ans. APC is the ratio of total consumption to total income. APC = C/Y.

10. Define average propensity to save.
Ans. Average propensity to save is the ratio of total saving to total income. APS=S/Y.

11. Define planned saving (or ex-ante saving).
Ans. The savers in an economy intend or plan to save is called planned (or ex-ante) saving.

12. Define planned investment.
Ans. The investors in an economy intend or plan to invest is called planned (or desired) investment.

13. Define multiplier.
Ans. Multiplier means the ratio of change in income to the change in investment. $K=\triangle Y/\triangle I$

14. What is meant by saving-function?
Ans. Saving-function means a functional relationship between total saving and total income.

$$S = f(Y)$$

15. Explain the meaning of consumption-function.
Ans. Consumption-function means a functional relationship between total consumption and total disposable income.

$$C = f(Y)$$

16. What do you understand by break-even point?

Ans. Break-even point indicates a point where consumption becomes equal to income or consumption curve cuts the income curve.

17. Mention the relation of investment multiplier to MPC and MPS.
Ans. There is direct and positive relationship between multiplier and MPC. There is negative relationship between multiplier and MPS.

18. Tell the meaning of average propensity to save. Can the value of APS be negative?
Ans. APS is the ratio of total saving and total income (APS=S/Y). APS can be negative when consumption expenditure is greater than income.

19. What can be the minimum and the maximum values of multiplier?
Ans. The minimum value of multiplier can be one and its maximum value can be infinity.

20. Can the value of average propensity to save be negative? If yes, when?
Ans. APS becomes negative when total consumption is greater than income.

21. What is the relationship between marginal propensity to save and multiplier?
Ans. There is negative relationship between MPS and multiplier.

22. Give the meaning of autonomous consumption.
Ans. The consumption which is done at zero level of income is known as autonomous consumption.

23. State the law of consumption.
Ans. The law of consumption states that as income increases, consumption also increases but less than the increase in income.

24. When can a country achieve full employment equilibrium level of output and income?
Ans. When actual aggregate demand is equal to the aggregate demand required for full employment.

25. What is deficient demand? Define it.
Ans. Deficient demand refers to a situation where actual aggregate demand falls short of the aggregate demand required for full employment. AD <AD.

26. What is meant by excess demand in macroeconomics ?
Ans. When actual aggregate demand becomes higher than the amount of aggregate demand required for full employment, it is called excess demand. AD > ADf

27. Define deflationary gap.
Ans. The extent to which actual aggregate demand falls short of the aggregate demand required for full employment, is termed as deflationary gap.

28. What is inflationary gap? Define it.

Ans. The extent to which actual aggregate for demand becomes higher than the aggregate demand required for full employment is referred to as inflationary gap.

29. Is under-employment equilibrium possible? When does it happen?
Ans. Yes, under-employment equilibrium is possible. It happens when actual aggregate demand falls short of the aggregate demand required for full employment.

30. When actual aggregate demand falls short of the aggregate demand required for full employment level, what do you call it?
Ans. Deficient demand or deflationary gap.

31. Give meaning of involuntary unemployment or define involuntary unemployment.
Ans. Involuntary unemployment refers to the situation when workers are prepared to work at current or slightly lower wage rate, but jobs are not available to them due to deficiency of aggregate demand.

32. What is meant by full employment equilibrium?
Ans. When aggregate demand equals to the aggregate supply at a point where all the resources of the country get employment, this is called full employment equilibrium.

33. What is meant by under-employment equilibrium?
Ans. When the equilibrium between aggregate demand and 45° line (or aggregate supply) takes place at a less-than-full employment point, it is termed as under-employment equilibrium.

34. Define the following:
(i) Voluntary unemployment.
(ii) Involuntary unemployment.
Ans. (i) When workers are not prepared to work at current wage rate, it is termed as a situation of voluntary unemployment.
(ii) Involuntary unemployment refers to the situation when workers are prepared to work at current or slightly lower wage rate, but jobs are not available to them due to deficiency of aggregate demand.

35. What is ex-ante aggregate demand?
Ans. Ex-ante aggregate demand refers to the total planned expenditure on purchase of goods and services by an economy.

3-4 MARKS
QUESTIONS

1. Show the relationship between APC and APS.
Ans. Relationship between APS:
APC = C/Y and APS=S/Y
APC+APS = C/Y+S/Y $=\frac{C+S}{Y}=\frac{Y}{Y}=1$
Therefore,

APC+APS = 1
1- APC = APS
1- APS = APC

2. What is the relation between MPC and MPS?
Ans. Relationship between MPC and MPS:

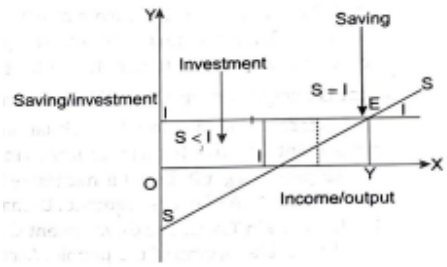
$$MPC = \frac{\triangle C}{\triangle Y} \text{ and } MPS = \frac{\triangle S}{\triangle Y}$$

$$MPC + MPS = \frac{\triangle C + \triangle S}{\triangle Y} = \frac{\triangle Y}{\triangle Y} = 1$$

Therefore,

MPC+MPS = 1
MPS= 1-MPC
MPC= 1-MPS

3. Explain with the help Of a diagram how equilibrium level of income in an economy is determined by saving and investment curves? Will there always full employment at equilibrium level of income?
Ans. Equilibrium level Of Income is determined at a where exante or planned saving is equal to planned investment. In the adjacent figure SS curve is saving curve and curve is investment Curve. Equilibrium Of income OY occurs at point E, Where Saving (SS) curve and investment curve (11) intersect. There may Or may not full employment at equilibrium level of income



4. State briefly the effect of excess demand on output, employment and price.
Ans. (i) Effect on Output: In the situation Of excess demand, output does not increase, The economy is already in the State Of full capacity production.
(ii) Effect on Employment: Employment will not increase there is no involuntary unemployment in the economy. The economy has already achieved full employment.
(iii) Effect on Prices: Excess demand only generates pressure of demand on the existing flow of goods and services, in the economy. Accordingly, prices tend to rise.

5. What is deficient demand in an economy? What is its impact on output. employment and prices?
Ans. Deficient demand refers to the Situation when aggregate demand (AD) is short Of aggregate supply (AS) corresponding to full employment in an economy.
(i)Effect on Output: Low level Of investment and employment implies low level Of Output.
(ii)Effect on Employment: Because Of deficiency Of demand investment level is reduced. Accordingly, level Of employment tends to fall.
(iii) Meet on Prices: Fall in prices is the immediate consequence Of deficient demand,

6. What do you mean by Aggregate supply? Give its components,

Ans. It refers to the money value of goods and services which producers are willing to supply in an economy in a given time Rtiod. AS is represented bv national income. It means AS and national income both are same, AS is determined by:
(i) level of employment Of resources.
(ii) level Of technology.
Components of Aggregate supply are consumption and savings. Thus, AS = C + S.

7• Explain the working Of investment multiplier with the help of numerical example.
Ans. Investment multiplier shows a relationship between initial increment in investment and the resulting increment in national income. Multiplier thus is a ratio of increase in national income due to an increase in investment. Symbolically $\frac{\triangle Y}{\triangle I}$. Investment multiplier indicates the multiplying effect of investment on income. The working of the multiplier can explained in the ways given ahead:
Change in investment causes changes in income. As a result there is a change in consumption. Consumption expenditure of one is an income of the other, Hence, change in consumption leads to change in income. This process continues till change in consumption falls to zero, Let us understand this with the help of an example:
(i) Suppose an additional investment of ₹100crores is made to construct a road. This extra investment will generate an extra income Of ₹100 crores.
(ii) If MPC is assumed to be 0.90 then recipient Of this additional income will spend 90% of ₹100 Crores, i.e.. ₹ 90 Crores on consumption expenditure and the remaining amount will be saved, It will increase the income by ₹90 crores.
(iii) NOW, 90 %of ₹90 crores, i.e., ₹81 crores Will be spent on consumption and the remaining amount Will be saved .
The multiplier process will go on and the consumption expenditure in every round will be O.90 times of the additional income received from the previous round.

8. What is inflationary gap? What is its impact on output and prices?
Ans. Inflationary gap is the excess of AD over and above its level required to maintain full employment equilibrium in the economy. Inflationary gap generates extra pressure on the existing now of g'XXis and services at the level Of full employment Accordingly, prices tend to rise. The output will not increase.

9. What is excess demand? Explain any two fiscal measures to excess demand.
Ans. Excess demand refers to a situation in which aggregate demand exceeds aggregate supply to full employment. The two fiscal measures to reduce excess demand are as follows:
(i) Surplus Budget Policy: It is necessary that Government's expenditure should be less than its income in order to correct the situation of excess demand. Demand will decrease following decrease in expenditure.
(ii) Increase in Taxes: The Government should levy new taxes and enhance the rate of the existing ones. This would reduce disposable income of the people. Accordingly aggregate demand will shrink.

10. State briefly the measures to correct excess demand.
Ans. The problem Of excess demand can solved by reducing aggregate demand as follows:
(i) The public expenditure is to be reduced.
(ii) The taxes are to be increased.
(iii) The bank rate is to be increased.

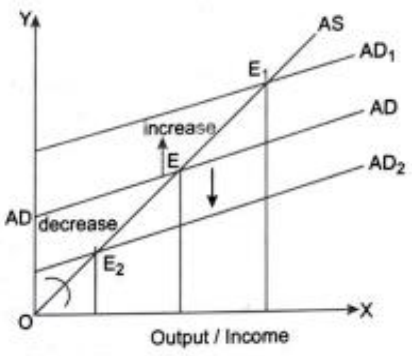
(iv) Securities should be sold in the open market.
(v) The availability and supply of money should be reduced.
(vi) The imports are to be increased.

11. Discuss any three measures by which deficient demand in an economy can corrected.
Ans. The problem of deficient demand can be solved by increasing aggregate demand as follows:
(i) Public expenditure on public works, public welfare and public investment should be increased.

(ii) The direct and indirect taxes should reduced, It will lead to increase in income and thereby, increase in demand.
(iii) The bank rate should be reduced, It will result in reducing rate of interest. Consequently, investment will increase.

12. Show the relationship multiplier & MPS and multiplier & MPC.
Ans. Multiplier and MPC: There is a direct relationship between MPC and multiplier. Higher the MPC more will be the value of multiplier and vice-versa. In case of higher MPC people will spend a large of their increased income on consumption. In such a way, value of multiplier will be more. On the other hand in case of low MPC, people will spend lesser proportion of their increased income on consumption. In such a case, value of multiplier will comparatively Iow $k=\frac{1}{1-MPC}$.
Multiplier and MPS : There is a negative and inverse relationship between MPS & multiplier, Greater the MPS smaller Will be the value of multiplier, Thus, $K = \frac{1}{MPS}$.

13. Show the effect of leakages and injections aggregate demand and aggregate supply
Ans. Shift in Equilibrium: of injections and withdrawals:
Withdrawals are the leakages in the circular flow of income which decreases AD.
Injections are the done in the economy which increases the flow of income and AD also.
But Leakages and injections do not have any effect on AS.



6-8 MARKS
QUESTIONS

1. Explain the components of aggregate demand.
Ans. Aggregate demand refers to the total amount of money which all the sections of the economy are ready to spend on purchase of goods and services produced in an economy during a period of one accounting year. AD is a synonym with aggregate expenditure.
Components of Aggregate Demand:
(i) Household Or Private Consumption Demand: This refers to the total expenditure incurred by all the households on purchase of goods and services for their personal consumption. Households spend their income on purchase of durables, semi-durables, services, perishable goods, etc. Such demand is influenced by many

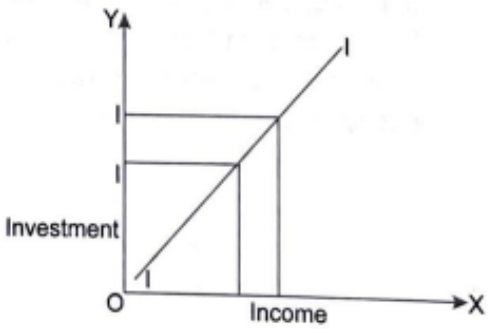
factors such as price of the commodity, income of the consumer and so on. Generally it depends on the disposable income of the economy. Disposable income refers to the income from all sources available to households for consumption and savings. Thus, higher the level of disposable income higher will be the private consumption expenditure and vice-versa.

(ii) Private Investment Expenditure: It refers to the total expenditure incurred by all private individuals on capital goods, i.e., the expenditure on creation of new capital assets like machines, raw material, etc. It is also known as capital formation.

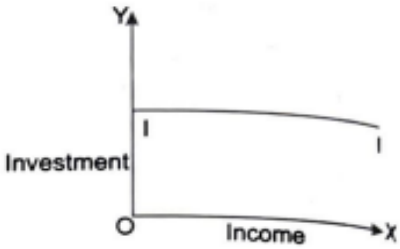
(iii) Government Expenditure: It refers to the total expenditure of the Government on consumer and capital goods to satisfy the common needs of the economy. Government expenditure primarily depends on the objective of social welfare.

(iv) Net Exports: It is a difference between exports of goods and imports during a given period of time.
Thus, AD = C+I+G+(X-M). But for simplifying the analysis, Keynes included only two major components namely consumption demand and investment demand. To put in symbols AD = C + I.

2. Explain induced and autonomous investment with the help of the graphs.
Ans. Investment: Investment means expenditure on creation of new capital assets. It includes the expenditure done on assets like machines, buildings, etc. There are two types of investments:
(i) Induced Investment: It refers to the investment which is made with the motive of earning profit as is done in private sector. It depends directly upon profit expectations. It is income elastic, it means if income goes up, induced investment also goes up and vice-versa.
The induced investment is the increasing function of profit. If firms expect profit, they are induced to invest. The profit expectation of firms depend upon aggregate demand for goods and services in the economy. The level of 'aggregate demand' itself depends upon the level of national income. The higher the level of national income, the higher thus is the level of induced Investment.



(ii) Autonomous Investment: The investment which is not influenced by changes. in national income is called autonomous investment. In other words, an autonomous investment is independent of the level of national income. As regards the size of autonomous investment, it is influenced by many basic factors such as increase in population, manpower, level of technology, the role of interest, the expectations of future economic growth and the role of capacity utilisation, etc.



3. Explain the psychological law of consumption. Also write down its assumption.
Ans. According to Keynes's fundamental psychological law, individuals are likely to make higher consumption as their income increases but not as much as the increase in their income. This law is also known as propensity to consumer consumption function. This law is based on following propositions:
Consumption tends to increase with every increase in income. Income may be zero, but consumption is always positive.

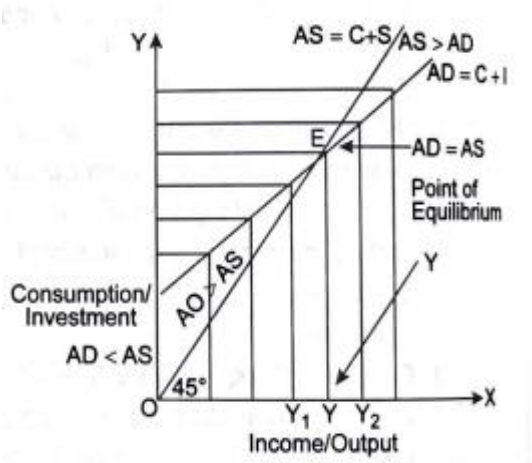
(i) When aggregate income increases, consumption expenditure also increases but by a smaller amount.
(ii) When income increases, the increment of income will be divided in certain proportion between consumption spending and saving.
(iii) As income increases, both consumption and saving will go up.

Assumptions of the law given by Keynes:

(i) Habits of the people regarding spending do not change or propensity to consume remains same.
(ii) Conditions are normal in the economic system. There is no war, revolution, hyperinflation, etc.
(iii) Existence of capitalistic economy behaviour of consumption.

4. Explain the theory of determination of income and employment with the help of aggregate demand and aggregate supply curves.

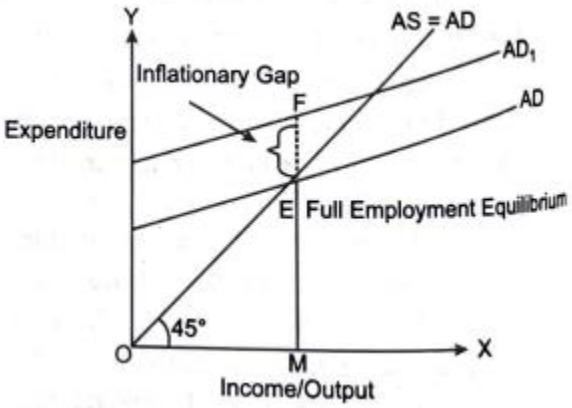
Ans. According to modern, theory of income and employment determination, in any economy at any given time, income and employment
are determined at that level where aggregate demand is equal to aggregate
supply.



In the adjacent figure AD represents aggregate demand curve and 45° line is the line of reference where AS = AD. Equilibrium level of income Y is determined at point E, where AD and AS are equal. Prior to point E, aggregate demand will exceed aggregate supply leading to an increase in level of income upto point E. Beyond point E, aggregate supply will exceed aggregate demand leading to a fall in income back toward point E.

5. With the help of a diagram explain the concept of inflationary gap.

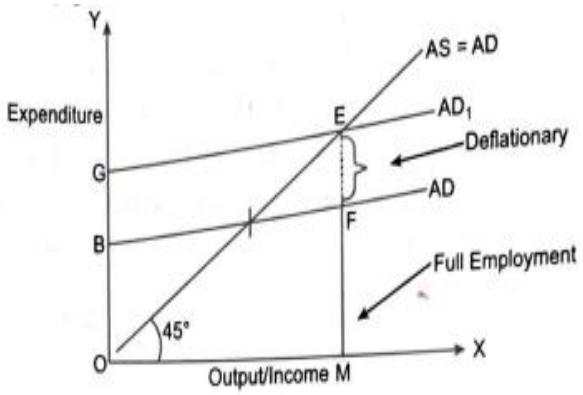
Ans. Inflationary gap is the excess of aggregate demand over and above its level required to maintain full employment equilibrium in the economy.



In the adjacent figure full employment equilibrium is at point E. If the level of demand increase to AD_1, it is in excess of what is required to maintain full employment. This causes inflation. Hence, the difference between AD1 and AD(EF) is called inflationary gap.

6. With the help of a diagram explain the concept of deflationary gap.

Ans. Deflationary gap is the deficiency of AD required to maintain full employment equilibrium. In the adjacent figure full employment equilibrium is at point E. However, if the level of demand happens to be AD, it is short of its full employment level. Accordingly. deflationary gap occurs to the tune of AD-AD = EF.



7. Give the meaning of excess demand in macroeconomics. Explain any four monetary policy measures to correct it.

Ans. Excess demand refers to the situation when aggregate demand (AD) is in excess of aggregate supply (AS) corresponding to full employment in the economy.

The following four monetary measures may be adopted to correct it:

(i) Increase in Bank Rate: Bank rate is increased to control excess demand. Increased in bank rate increases the rate of interest and credit becomes dear. Accordingly, demand for credit is reduced.

(ii) Open Market Operations: The Central Bank starts selling securities in open market so that some purchasing power is withdrawn from the banks.

(iii) Increase in Cash Reserve Ratio: Cash reserve ratio of the banks is enhanced with a view to restrict the flow of credit.

(iv) Increase in Statutory Liquidity Ratio: Statutory liquidity ratio is enhanced to restrict the lending by the banks.

8. Explain the causes of excess demand.

Ans. Causes of Excess Demand:

Inflationary gap is caused by the following factors:

(i) Increase in Money Supply: The first major cause of excess demand is increased in the supply of money which leads to increase in aggregate demand. Unless increase in aggregate demand (caused by increase in money supply) is offset by a similar increase in supply of goods and services it will generate an upward pressure on prices. Cheap monetary policy or the policy of credit expansion adopted by the monetary authority also leads to increase in the money supply which raises the demand for goods and services in the economy.

(ii) Increase in Disposable Income: When the disposable income of the people increases, it raises their demand for goods and services. Disposable income may increase with the rise in national income or reduction in taxes or reduction in the saving of the people.

(iii) Increase in Public Expenditure: Government activities have been increasing on a large scale. Consequently, Government expenditure has also been increasing at a fast rate. It has to spend a huge amount of money for meeting requirements of defense and for its various programmes for the country's economic development. This has raised aggregate demand for goods and services.

(iv) Deficit Financing: Deficit financing is taken to meet the excess of the Government expenditure over its revenue. This deficit is financed by printing more notes. Deficit financing, therefore leads to increase in money supply, and hence, is responsible for the price rise.

9. Explain the causes of deficient demand.

Ans. (i) Reduction in Private Consumption Expenditure: It is an important component of AD. Reduction in private consumption expenditure causes a series deficiency in AD. Private consumption expenditure may reduce owing to several reasons. However, the most important is reduction I propensity to consume or increase in propensity to save.

(ii) Reduction in Investment Expenditure: Investment expenditure is the other important component of AD. Any reduction in investment implies a reduction in AD. It may reduce in situations of poor business sentiments. It happened in all market economics of the world in the year 2009.

(ii) Reduction in Government Expenditure: Government expenditure is of vital significance in welfare economics. However, there could be situations of perpetual losses in public enterprises as in India in the recent past.

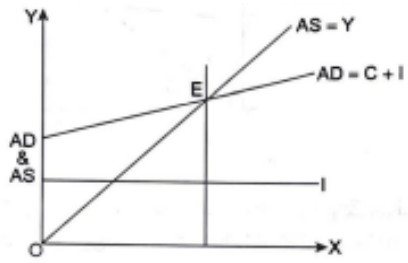
(a) Decline in Exports: Decline in exports may occur due to lack of demand in rest of the world.

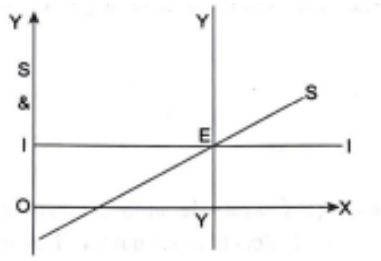
(b) Rise in Imports: Imports may rise when international prices are lower than the domestic prices. A rise in imports implies cut in AD.

(c) Increase in Tax Rates: Increase in tax rates leaves lesser disposable income with the people. It reduces their capacity to spend even when their propensity to spend remains the same. Lower disposable income means lower level of AD,

10. Show the simultaneous equality between AD & AS and S & I diagrammatically.

Ans. Simultaneous equality between AS & AD as well as between 5 & 1:





QUESTIONS BASED ON DIFFERENTIATION

1. Differentiate between APC and MPC.

Ans:

S.NO.	Basis	APC	MPC
1	**Meaning**	It refers to the ratio of consumption expenditure to total income at a point of time.	It refers to the ratio of change in consumption to the change in income at a point of time.
2	**Response to change in income**	When income increases APC falls but at a rate less than that of MPC.	When income increases MPC falls but at a rate more than that of APC.
3	**Formula**	APC = C/Y	MPC = ΔC/ ΔY

2. Differentiate between APS and MPS.

Ans.

S.NO.	Basis	APC	MPC
1	**Meaning**	It refers to the ratio of savings to the corresponding level of income at a point of time.	It refers to the ratio of change in savings to the change in total income over a period of time. MPS = AS / AY
2	**Formula**	APS = S / Y.	MPS = AS / AY.
3	**Value less than 0**	APS can be less than 0 when consumption is more than national income.	MPS can never be less than 0 as change in saving can never be negative, as change in consumption can never be more than change in income.

3. Differentiate between induced and autonomous investment.

Ans.

S.NO.	Basic	Induced Investment	Autonomous Investment
1	**Aim**	It is driven by profit motive.	It is done for social welfare not for profit.
2	**Income elastic**	It is income elastic, je., increases with increase in income and vice-versa.	It is unaffected by the changes in the level of income.
3	**Shape of the curve**	Its curve slopes upward as it has direct relations with the income.	Its curve is parallel to the X-axis as it is income inelastic.
4	**Sectors**	It is generally done by private sector.	It is generally done by Government sector.
5	**Determining factors**	Besides income it also depends on innovations, technological changes, composition of population, etc.	It depends on socio-economic and political conditions of the economy.

4. Differentiate between inflationary gap and deflationary gap.

Ans:

S.NO.	Basic	Inflationary Gap	Deflationary Gap
1	**Meaning**	It refers to the gap by which actual AD exceeds AS at level of full employment	It refers to the gap by which actual AD falls as a level of full employment.
2	**Impact**	It does not affect the output and employment.	It leads to fall in output and employment due to deficiency in aggregate demand.
3	**Equilibrium**	It result in rise in general price level.	It result in fall in general price level.

5. Distinguish between excess demand and deficient demand.

Ans.

S.NO.	Basic	Excess Demand	Deficient Demand
1	**Meaning**	It refers to the situation when AD is more than AS corresponding to full employment level in the economy.	It refers to the situation when AD is less than AS correspondence to full employment level in the economy.
2	**Impact**	It causes inflation.	It causes deflation.
3	**Equilibrium**	It indicates over full employment equilibrium.	It indicates under employment equilibrium.

CASE BASED QUESTIONS

CASE 1

According to Keynes' own theory of income and employment: "In the short period, level of national income and so of employment is determined by aggregate demand and aggregate supply in the country. The equilibrium of national income occurs where aggregate demand is equal to aggregate supply. This equilibrium is also called effective demand point".

Answer the following questions:

1. What do you mean by Aggregate demand?
2. What are the components of aggregate demand?
3. What is Aggregate supply?

Ans. 1. Aggregate demand is a measurement of the total amount of demand for all finished goods and services produced in an economy. Aggregate demand is expressed as the total amount of money exchanged for those goods and services at a specific price level and point in time.

2. There are four main components of aggregate demand. They are consumption, investment, government spending and net exports (exports minus imports).

3. Aggregate supply is the total quantity of output firms will produce and sell-in other words, the real GDP. The upward- sloping aggregate supply curve-also known as the short run aggregate supply curve-shows the positive relationship between price level and real GDP in the short run.

CASE 2

Investment multiplier is an important part of economic theories suggested by notable economist John Maynard Keynes. According to this concept, in the event of an increase in the investment activities either public or private which can be in the form of private consumption spending, government spending in an economy, there is a corresponding increase in the Gross Domestic Product (GDP) of the economy by a value more than the amount invested.

In simple words, investment multiplier refers to the increase in the aggregate income of the economy as a result of an increase in the investments done by the government in the form of new projects.

The size of the investment multiplier is determined by the decisions of the households in an economy in the areas of spending or saving.

Answer the following questions:

1. Define investment multiplier and give the formula to calculate it.
2. What is the relation between MPC and Multiplier?
3. How is MPS related to the Multiplier?

Ans. 1. Investment multiplier shows a relationship between initial increment in investment and the resulting increment in national income. Multiplier thus is a ratio of increase in national income due to an increase in investment. Symbolically K=AY/AL.

2. Relationship between multiplier and MPC, since K = 1/1 - MPC so the value of multiplier varies directly with the value of MPC. Higher the value of MPC the larger will be the value of multiplier and lower the value of MPC the smaller will be the value of multiplier.

3. There is an inverse relationship between investment multiplier and marginal propensity to save which means if marginal propensity to save increases, investment multiplier decreases and vice-versa.

CASE 3

Investment means acquiring the assets that will give returns on the Invested amount after a certain period of time. The investment is made with the goods or services, it is not consuming today but it will be used in the future to create wealth, The main goal of the investment is to generate more income or increase wealth. There are many types of Investment in the Economy. That helps the country to grow the economy. Some of the important types of investment are: (1) Business Fixed Investment, (2) Residential Investment, (3) Inventory Investment, (4) Autonomous Investment, and (5) Induced Investment.

Answer the following questions:

1. What is an autonomous investment?
2. Differentiate between autonomous and induced investment.
3. Autonomous investment is income.

(i) Elastic
(ii) Inelastic
(iii) Dependent
(iv) None of the above

Ans. 1. Autonomous investment is the portion of the total investment made by a government or other institution independent of economic considerations.

2. Induced investment is that investment which is governed by income and amount of profit in return, i.e., higher profit may lead to higher investment and vice-versa. Autonomous investment is that investment which is independent of the level of income or profit and is not induced by any changes in the income.

3. (ii) Inelastic

QUESTIONS WITH HIGH DIFFICULTY LEVEL

1. Savings and Investments are always equal. Discuss.
Ans. S and I can be realised (ex-post) or planned (ex-ante). Planned S and planned I are equal only at equilibrium level.
Realised S and realised I are always equal. The equality is derived as follows:

$$Y=C+S$$
$$Y=C+1$$

Hence, $S=1$

2. Can consumption be greater than income? When does it happen? What is the saving in this situation?
Ans. Yes, it happens when income is less than the minimum consumption level. Negative saving.

3. Why the value of APC can never be negative?
Ans. The value of APC can never be negative because the level of total consumption cannot be negative at any level of income.

4. When does MPS become zero?
Ans. When MPC-1 then MPS becomes zero.

5. In which situation multiplier can be infinity?
Ans. Multiplier can be infinity only when MPC is equal to 1.

6. If MPC and MPS are equal, what is the value of the multiplier?
Ans. Since, MPC + MPS = 1 and if MPC = MPS, therefore the value of MPS = $\frac{1}{2}$ and value of multiplier is 2.

7. State the direction of change in income and output, when: (i) AD> output, (ii) AD < output, (iii) AD = output, (iv) S>L,(v) S<L, (vi) S=1.
Ans. (i) Increase, (ii) Decrease, (iii) Equilibrium, (iv) Decrease, (v) Increase, (vi) Equilibrium.

8. What is the value of MPC when MPS is zero?
Ans. MPC-1 (Because MPC=1-MPS=1-0).

9. Why can the value of MPC not be greater than one?
Ans. Because increment in consumption remains less than the corresponding increment in income.

10. What will be the impact of the following on aggregate demand?
(i) Increase or decrease in public expenditure.
(ii) Increase or decrease in taxes.
(iii) Increase or decrease in reserve ratio.
(iv) Increase or decrease in interest rate.

Ans. (i) Aggregate demand increases with the increase in public expenditure and decreases with the decrease in public expenditure.
(ii) Increase in taxes reduces aggregate demand and decrease in taxes raises the level of aggregate demand.
(iii) Aggregate demand decreases with the increase in reserve ratio and increases with the decrease in reserve ratio.
(iv) Aggregate demand decreases with the increase in interest rate and increases with the decrease in interest rate.

11. What is natural rate of unemployment?
Ans. It refers to unemployment which is bound to exist even when labour market is in a state of equilibrium. It occurs largely due to:
(i) time required in shifting from one job to the other and
(ii) time required in adjusting to change in technology.

Numericals

Important Formulae:
Y=C+S,
AD = C + I,
AS = C+S,
APC = C/Y = 1 – APS,
APS = S/Y = 1 - APC,
MPC = ΔC/ΔY=1-MPS,
MPS = ΔS/ ΔY = 1- MPC,
K=ΔY/ΔI=1/1-MPC=1/MPS.

1. How much is the MPS in an economy in which MPC is 0.8 and 0.75?
Ans. MPS = 1-MPC
MPS = 1 - 0.8 = 0.2
MPS = 1 - 0.75 = 0.25.

2. How much is the MPC in an economy in which MPS is 0.2?
Ans. MPC = 1-MPS
MPC = 1-0.2 = 0.8.

3. If APS is 0.6, how much will be APC?
Ans. APC = 1 - 0.6 = 0.4.

4. If APC is 0.7, how much will be APS?
Ans. APS = 1 - 0.7 = 0.3

5. If disposable income is ₹ 500 and saving is ₹ 100, find out APC.
Ans. APC = 400 (500 - 100)/500 = 4/5 = 0.8.

6. If MPS is 1, how much will be MPC?
Ans. MPC = 1 - MPS = 1-1 = 0.

7. If APC is 0.75, how much will be APS?
Ans. APS = 1 - APC = 1-0.75 = 0.25.

8. In an economy investment increases by ₹ 10 crores. As a result income increases by ₹ 50 crores. What is the value of multiplier?
Ans. K=50/10 = 5.

9. What will be the value of multiplier if MPS is 0.4?
Ans. Multiplier (K) = 1/MPS = 1/0.4 = 10/4 = 2.5.

10. If the value of MPS is 0.25, what is the value of multiplier (K)?
Ans. K = 1 / MPS = 1 / 0.25 = $\frac{1}{1/4}$ = 4

11. Complete the following table:

Income	Consumption	MPC	MPS
800	750	-	-
1000	930	-	-
1200	1100	-	-
1400	1260	-	-

Ans:

Income	Consumption	MPC= $\Delta C/\Delta Y$	MPS= $\Delta S/\Delta Y$	Saving = Y-C
800	750	-	-	50
1000	930	0.9	0.1	70
1200	1100	0.15	0.15	100
1400	1260	0.2	0.2	140

12. Complete the following table:

Level of Income	Consumption Expenditure	Marginal Propensity to consume	Marginal Propensity to Save
400	240	-	-
500	320	-	-
600	395	-	-
700	465	-	-

Ans:

Level of Income	Consumption Expenditure	Marginal Propensity to consume	Marginal Propensity to Save
400	240	-	-
500	320	0.8	0.2
600	395	0.75	0.25
700	465	0.7	0.3

13. Complete the table given below:

Income	Saving	Marginal Propensity to consume	Marginal Propensity to Save
0	-20	-	-
50	-100	-	-
100	0	-	-
150	30		
200	60	-	-

Ans:

Income	Saving	Marginal Propensity to consume	Marginal Propensity to Save
0	-20	-	-
50	-100	1.4	1.6
100	0	1	-2
150	30	0.4	0.6
200	60	0.4	0.6

14. In an economy, a 40 percent increase in investment result in a 40 percent increase in income. Calculate the marginal propensity to consume.
Ans. MPC = 0.

15. Marginal propensity to consume is zero. Calculate the change in income if investment falls by ₹1000 crores.
Ans. ₹1000 crores.

WORKSHEET FOR SELF EVALUATION

1. What will be the impact of the following on aggregate demand: **(2 Marks)**
(i) Increase or decrease in public expenditure?
(ii) Increase or decrease in taxes?
(ii) Increase or decrease in reserve ratio?
(iv) Increase or decrease in interest rate?

2. Calculate APC and APS from the following schedule: **(2 Marks)**

Income (Y) (in ₹)	Consumption (C) (in₹)
100	80
200	120
300	180

3. From the following schedule, compute APC, APS, MPC and MPS: **(2 Marks)**

Income	200	250	300	350	400
Saving	-	5	15	20	50
APC					
APS					
MPC					
MPS					

4. Mention the components of aggregate supply. **(2 Marks)**
5. Can the value of APC be greater than one? **(2 Marks)**
6. Can APC be ever zero? **(2 Marks)**
7. Can the value of MPC be greater than one? **(2 Marks)**
8. What can be the maximum value of marginal propensity to save? **(2 Marks)**
9. Will ex-ante savings always be equal to the ex-ante investment? **(2 Marks)**
10. What happens to the level of national income when aggregate supply falls short of aggregate demand? **(2 Marks)**
11. What is meant by full employment equilibrium? **(2 Marks)**
12. How is multiplier related to MPC and MPS? **(2 Marks)**
13. If MPC and MPS are equal, what is the value of multiplier? **(2 Marks)**
14. What do you mean by Ex-ante saving and Ex-post saving? **(2 Marks)**
15. Differentiate between AD and AS on the basis of their meaning. **(2 Marks)**
16. What changes will take place to bring an economy in equilibrium: **(6 Marks)**
(i) If planned savings are more than planned investment?
(ii) If planned savings are less than planned investment?
(iii) If there is an increase in MPC what will be its impact on MPS?
17. If MPS = 0.2, what would be the value of MPC? **(2 Marks)**
18. What will be the magnitude of investment multiplier when MPC = 0.8? **(2 Marks)**
19. What will be the magnitude of investment multiplier when MPS = 0.5? **(2 Marks)**
20. As a result of an increase in autonomous investment by 125 crores, national income rises by 7 500 crores. Calculate the MPC. **(3 Marks)**